

Pronunciation Training through Articulation Motion Sensing

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ABSTRACT

Vowels are considered the essence of the syllable, which controls the articulation of each word uttered. However, articulation sensing has not been adequately evaluated. The challenging task is that the speech signal contains insufficient information for articulation analysis. It is difficult for users to improve their pronunciation only by getting scoring feedback on pronunciation assessments. We propose a new method to simultaneously use two different acoustic signals (speech and ultrasonic) to recognize lip shape and tongue position. The system gives articulation feedback to a user, identifying the articulation of monophthongs in multiple languages. The proposed technique is implemented into an off-the-shelf smartphone to be more accessible.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Natural language interfaces; • **Applied computing** → Education; Computer-assisted instruction.

KEYWORDS

Pronunciation Training, Acoustic Sensing, Ultrasonic Sensing, Acoustic Phonetics

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1 INTRODUCTION

Articulation problems produce varying degrees of pronunciation abnormalities and affect all aspects of people's lives, such as interpersonal communication, personality, adaptive social capacity, and learning ability. Moreover, it seriously reduces speech intelligence and speech communication. One crucial problem in the pronunciation learning context is that most systems cannot analyze articulation movement. The existing applications on the market

can only link signal correlation between native speakers and learners. Then, the applications send the scores to learners in a scale system or percentage without providing any feedback on correcting learners' articulation. However, many research pieces have studied the interaction between the user and application; we still need more mediums to calibrate the articulation and learners' pronunciation [1]. However, most of these applications focus only on accent training rather than pronunciation training, such as tone, time of each syllabus, and intonation, which may unintentionally mislead many foreign language learners by pronunciation training applications. Because the learners attempt to reach the highest score by trying to tune their pitch (Intonation) close to the examples rather than adjust their articulation concerning the pronunciation rules (Phonetics level) [1]. Furthermore, the impact of articulation detection benefits a wide range of applications, not limited to articulation disorder or language learning [2].

2 SYSTEM IMPLEMENTATION

In order to solve the problems mentioned above, we propose our prototype system to sense speech articulation, namely tongue and lip motion respecting monophthongs. The system employs two ranges of acoustic signals, which are speech and ultrasonic signals. It is designed to recognize monophthong articulation regarding tongue position and lip shape. Therefore, learners can adjust their wrong articulation with feedback. We take the phonetic rule as a priority task to map the monophthong vowels. The system can distinguish each vowel by tongue and lip according to the vowel diagram. The scheme applies to present a breakthrough method of pronunciation learning scenario. The system framework is shown in Figure 1. We divide the system into four parts: UI module, data acquisition module, tongue position detection module, lip motion detection module, and pronunciation evaluation module.

Data acquisition module: This module emits 20k Hz ultrasonic waves by the controlled smartphone's built-in speaker. And then, the phone's microphone receives the reflected audio signal for further processing and evaluation.

Tongue position detection module: This module uses linear predictive coding (LPC) to process the audio signal to estimate tongue height and represents tongue backness represented by Formant frequency. Finally, tongue height and backness can be mapped to the vowel diagram.

Lip motion detection module: This module analyzes the frequency shifting direction of the doppler effect from the generated ultrasonic signal at 20k Hz by comparing the lip movement peaks around 40 Hz or 19.96k Hz to 20.04k Hz center the central peak (~20k Hz).

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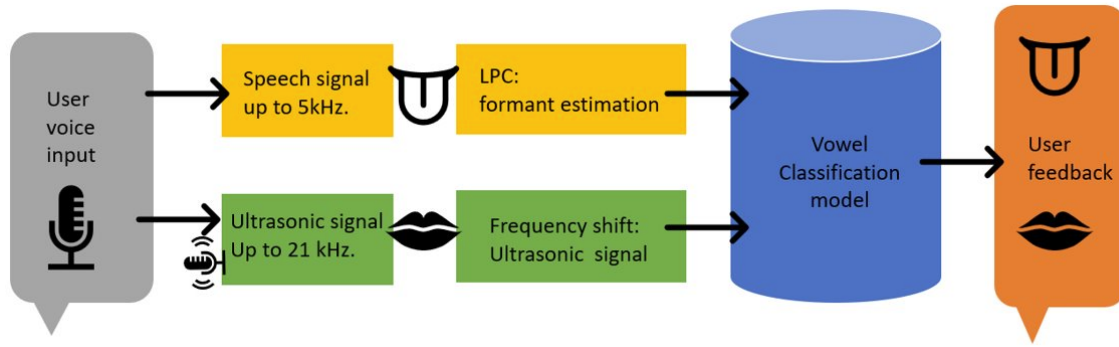


Figure 1: The system framework

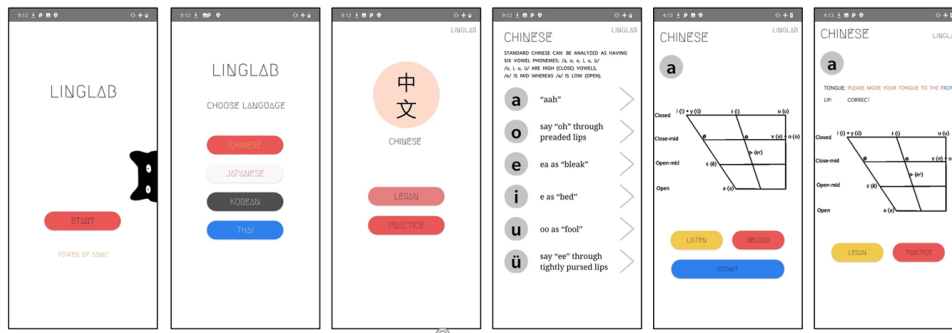


Figure 2: The prototype of the user interface

Pronunciation evaluation module: The user's pronunciation feedback is given from tongue position and lip shape input. Feedback is evaluated by combining the formant frequency and the direction of the frequency shift based on the pronunciation rule.

UI module: The UI module is used to write the system's user interface, providing a simple and easy-to-use interface for users who want to practice. There are seven Activities written in this system. First, users should select the vowel, e.g., /a/, which they want to practice. The LearnActivity and PracticeActivity are loaded in sequence. In PracticeActivity, users can record their pronunciation by clicking the record button, and the submit button to process the acoustic signal. Lastly, the user can jump to CorrectionActivity to check whether the tongue position and lip shape are correct. The prototype of the Android-based pronunciation training user interface is shown in Figure 2

3 RESULT AND EVALUATION

A prototype version proposed by our method was tested by twenty international students of the Chinese language over one week in our college. The learners are encouraged to be aware of pronunciation by self-adjusting their misplaced articulation. The evaluation was conducted to assess the effectiveness of learning Chinese pronunciation. We take the articulation of phonetic rule as a priority task to map the vowels—the histogram in figure 3 summarizes up the accuracy of their pronunciation before and after using the system. The pronunciation accuracy of 90% of the students improved with

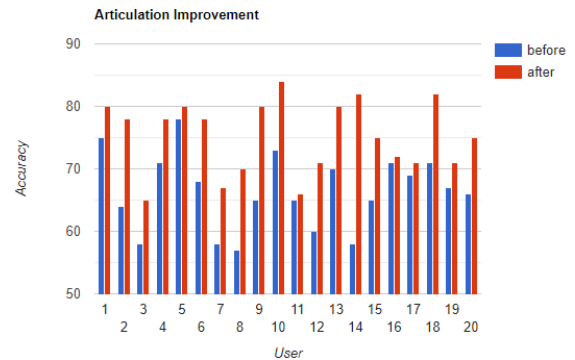


Figure 3: User improvement before and after using

a noticeable mean accuracy difference of 10.5. The remaining 10% of the students got a slight improvement for unknown reasons by an average of 2.68%. We notice an improvement by observing the dependent T-Test; There was a significant increase in the pronunciation in the week after using $M = 76.95$, $SD = 5.19$ compared to the week before ($M = 66.45$, $SD = 6.03$), $t = 11.39$, $p < 0$. We also collected the suggestions if the participants provide. Participants feel that they can easily engage in in-depth and effective interactions, which can help them explore the recommendation results. The visual design of the system seemed to be instructive and was easy to use for most participants. Some of them mentioned that

they gained insight into the usage of the UI module. Many of them agreed that the system could increase their motivation and enhance language learning. Consequently, they are willing to spend more time practicing pronunciation.

4 CONCLUSION

This paper presents a new pronunciation detection method based on the analysis of speech and ultrasonic acoustic signals. Using a smartphone to generate the ultrasonic signal, capture the reflections and analyze the fine-grained frequency shifting direction of articulation movements. Therefore, the system can analyze vowels by the tongue position and the lip shape. The system introduces a breakthrough approach for the pronunciation learning system, designed to recognize monophthong vowels' articulation regarding tongue position and lip shape without giving score feedback. Our goal is to explore generating visual graphic feedback of the wrong

articulation to the learner for better understanding and deeper engagement. The influence of foreign language learning and other types of vowels has excellent potential for further development. In future work, we also plan to develop another module that could be more practical by extracting the initials, finals, tones, and other speech organs in multiple languages.

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